IPPO: scale factor of the multi-scale maps (2, 3, 4)

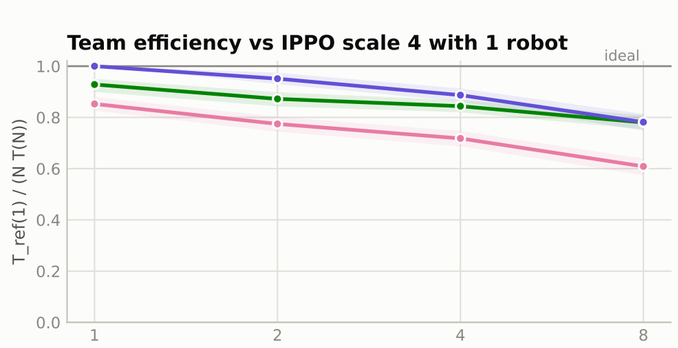
Same episodes for every policy (maps, start poses, seed), deterministic actions. Lines: mean over maps, band: 95 % bootstrap CI over maps.
Without flip_when_stuck. Team efficiency relative to IPPO scale 4 with 1 robot on the same map.

IPPO scale 2 IPPO scale 3 IPPO scale 4

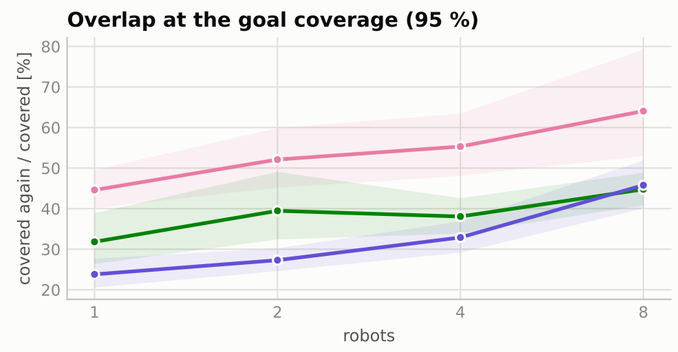
paper

38 maps x 3 episodes, goal 95 % coverage, limit 333 min

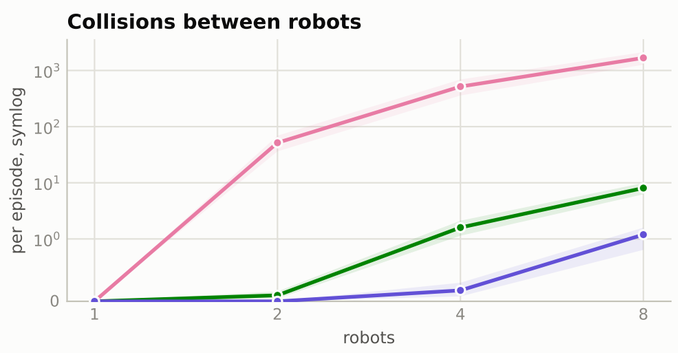
Success rate (95 %)



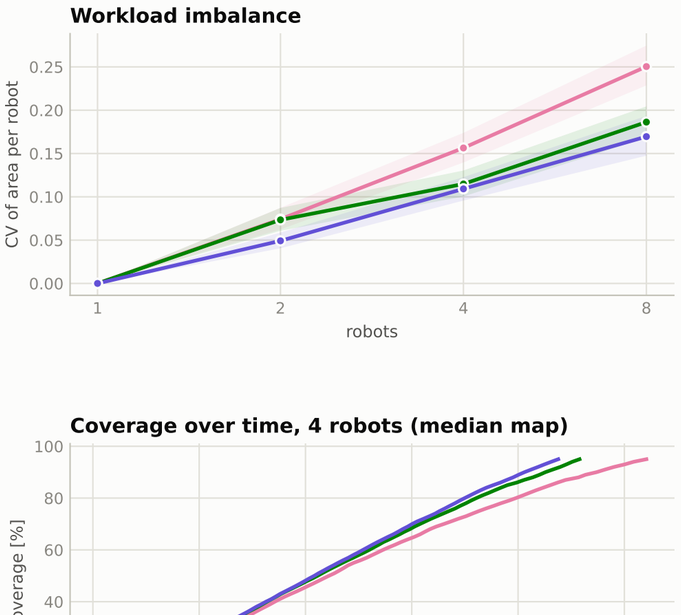
Time to the goal coverage (95 %)



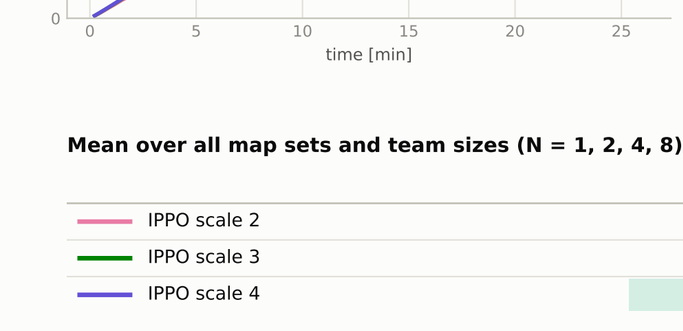
Team efficiency vs IPPO scale 4 with 1 robot



Overlap at the goal coverage (95 %)



Collisions between robots



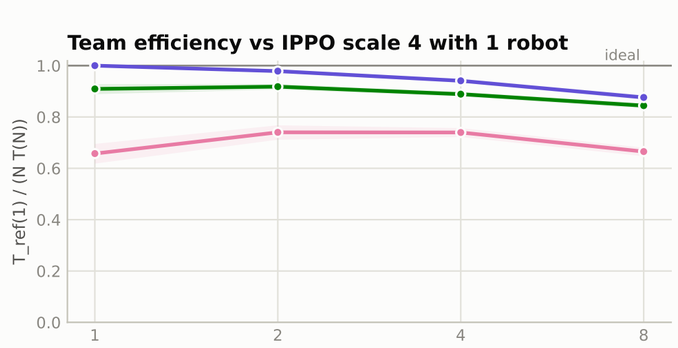
Workload imbalance

Coverage over time, 4 robots (median map)

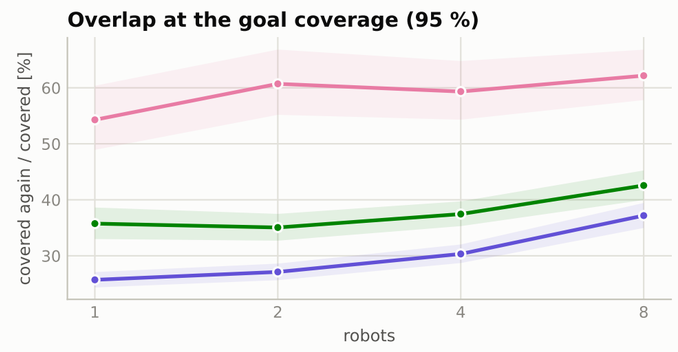
test_large

100 maps x 3 episodes, goal 95 % coverage, limit 250 min

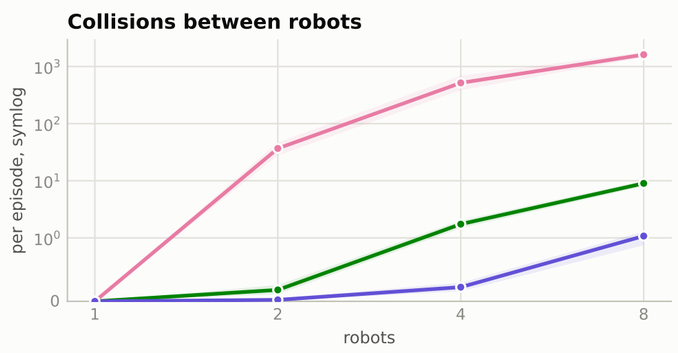
Success rate (95 %)



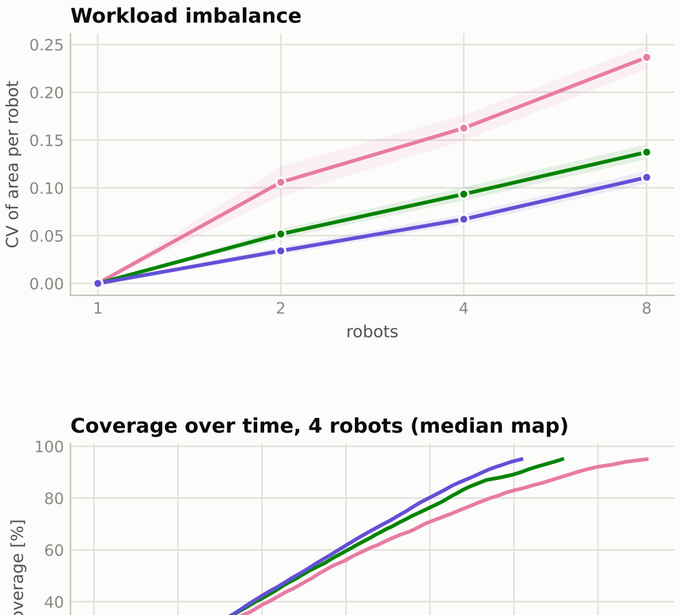
Time to the goal coverage (95 %)



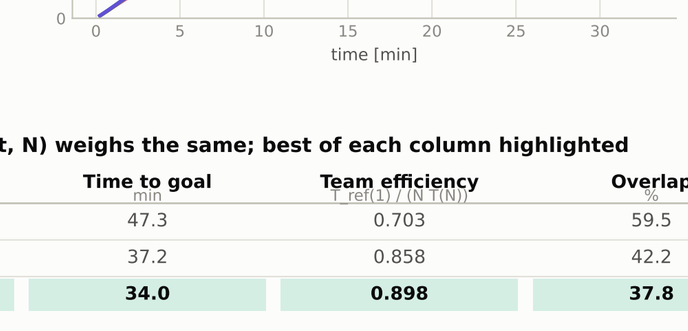
Team efficiency vs IPPO scale 4 with 1 robot



Overlap at the goal coverage (95 %)



Collisions between robots



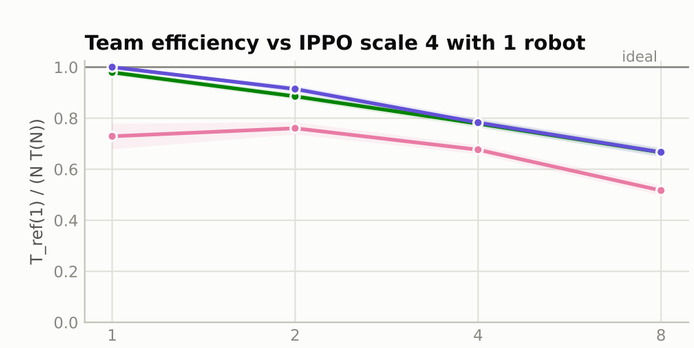
Workload imbalance

Coverage over time, 4 robots (median map)

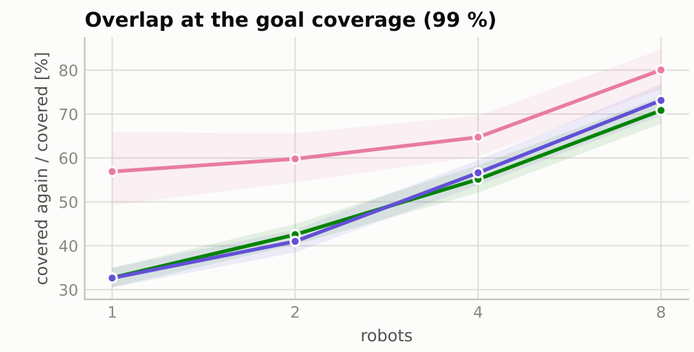
test_small

100 maps x 3 episodes, goal 99 % coverage, limit 67 min

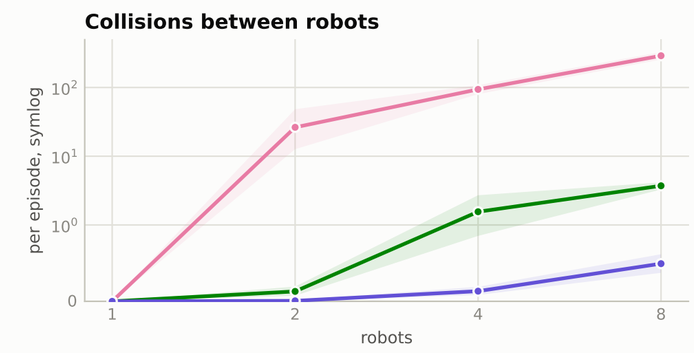
Success rate (99 %)



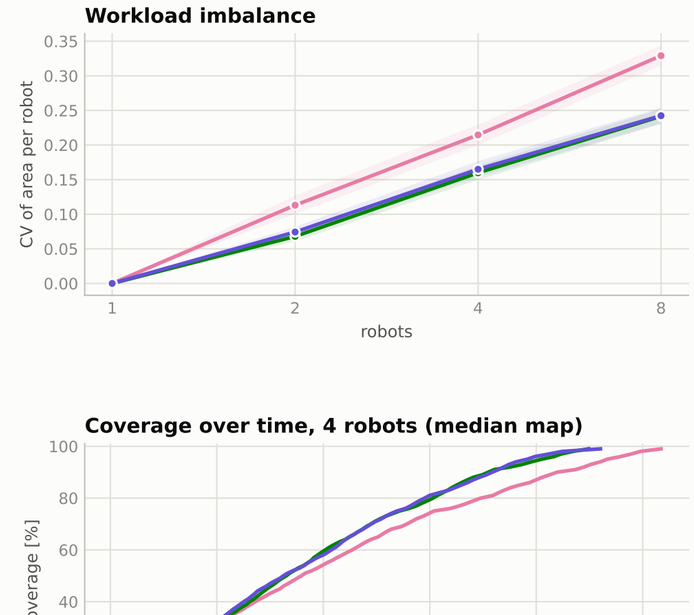
Time to the goal coverage (99 %)



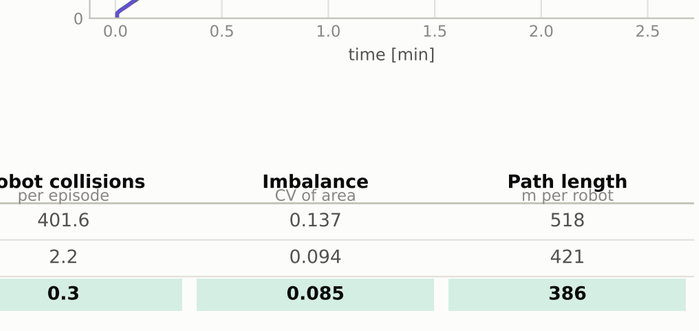
Team efficiency vs IPPO scale 4 with 1 robot



Overlap at the goal coverage (99 %)



Collisions between robots



Workload imbalance

Coverage over time, 4 robots (median map)

Mean over all map sets and team sizes (N = 1, 2, 4, 8), each (map set, N) weighs the same; best of each column highlighted

	Success %	Time to goal min	Team efficiency T_ref(1) / (N T(N))	Overlap %	Robot collisions per episode	Imbalance CV of area	Path length m per robot
IPPO scale 2	94.7	47.3	0.703	59.5	401.6	0.137	518
IPPO scale 3	98.8	37.2	0.858	42.2	2.2	0.094	421
IPPO scale 4	**99.5**	**34.0**	**0.898**	**37.8**	**0.3**	**0.085**	**386**